1. Problem Statement

Urban areas are experiencing a rapid increase in the number of vehicles, which has created significant challenges in **parking management**. Traditional parking systems are often manual and inefficient, requiring human supervision to record vehicle entry and exit times and to calculate parking fees. This approach leads to several problems, including **human errors, time delays, congestion at parking entrances, and a lack of accurate records**.

In many parking facilities, drivers must wait while attendants manually register vehicle information and determine the duration of parking. Additionally, the absence of automated tracking systems makes it difficult to monitor the **exact time a vehicle enters and leaves the parking area**, resulting in inaccurate billing and inefficient space utilization.

Therefore, there is a need for an **automated smart parking system** capable of identifying vehicles, recording their entry time, monitoring the duration of parking, and automatically calculating the payment required when the vehicle exits. Such a system would improve efficiency, reduce human intervention, and provide a more reliable and transparent parking management process.

2. Solution / Methodology

To address the limitations of traditional parking systems, this project proposes the design and implementation of a **Smart Parking Management System using a microcontroller-based platform**.

The proposed system operates by integrating sensors, a camera module, and a microcontroller to automate the process of vehicle identification, time tracking, and fee calculation.

When a vehicle arrives at the parking entrance, a distance sensor detects its presence and triggers the system to capture the **vehicle plate number** using a camera module. Image processing techniques are used to extract the plate number and transmit it to the microcontroller system. The system then records the **vehicle identification and entry time** using a real-time clock module and stores this information in the system memory.

While the vehicle remains in the parking facility, the system continuously maintains the stored record of the vehicle. When the vehicle reaches the exit gate, another sensor detects its presence and triggers the system to read the vehicle plate number again. The system retrieves the stored entry time corresponding to that plate number and calculates the **total parking duration**.

Based on predefined pricing rules, the system calculates the **amount to be paid** and displays it on a screen. After the payment is confirmed, the system removes the vehicle record from the database and opens the exit gate.

This methodology ensures automated vehicle tracking, accurate time measurement, and transparent payment calculation.

3. Technology / Prototype Requirements

The prototype of the smart parking system will be implemented using a combination of **embedded hardware components and software tools**.

Hardware Components

- ✓ **Arduino Uno** (Microcontroller)
  Acts as the central controller that processes sensor data and manages system operations.
- ✓ **ESP32-CAM Camera Module**
  Used to capture images of vehicle plate numbers for identification.
- ✓ **HC-SR04 Ultrasonic Sensor**
  Detects the presence of vehicles at the entrance and exit points.
- ✓ **DS3231 RTC Module**
  Provides accurate time tracking for recording vehicle entry and exit times.
- ✓ **16x2 LCD Display**
  Displays the plate number, parking duration, and payment amount.
- ✓ **SG90 Servo Motor**
  Controls the opening and closing of the parking gate.
- ✓ **Micro SD Card Module**
  Stores parking records such as plate numbers and entry times.

Software Tools

- ✓ **Arduino IDE**
  Used for programming and uploading the control code to the microcontroller.
- ✓ **OpenCV**
  Used for image processing and vehicle plate detection.
- ✓ **Tesseract OCR**
  Extracts text from captured license plate images.

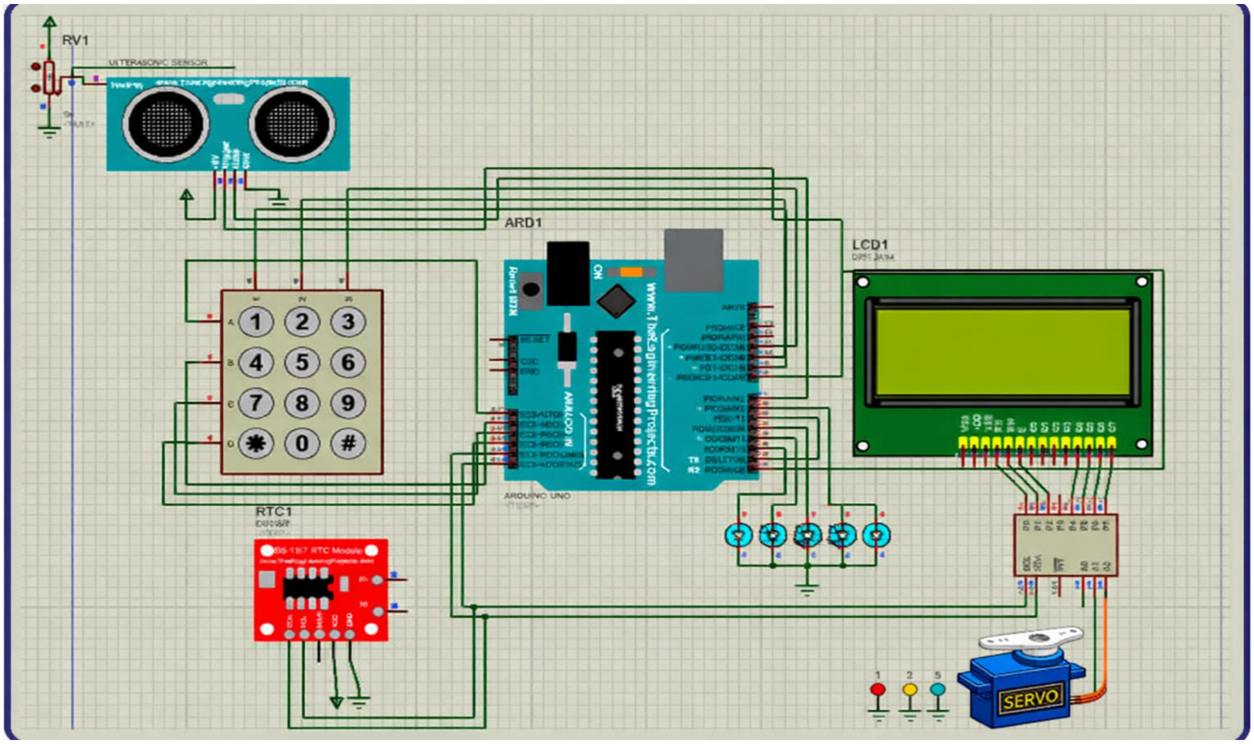

Figure: Automated smart park management system design prototype

4. Conclusion

The proposed Smart Parking System demonstrates how embedded systems and automation technologies can improve the efficiency and reliability of parking management. By integrating sensors, microcontrollers, and image processing techniques, the system enables automatic vehicle identification, accurate time tracking, and transparent parking fee calculation.

The implementation of this system can significantly reduce manual work, minimize human errors, and decrease waiting times at parking facilities. Furthermore, the system provides a scalable foundation that can be expanded with additional features such as mobile payment integration, real-time parking space monitoring, and cloud-based data management.

Overall, the project highlights the potential of **smart embedded solutions** to support the development of intelligent transportation infrastructure in modern cities.